235199: metadacite, Duketon

(Burtville Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Duketon (SG 51-14), Duketon (3342)
MGA Zone 51, 429788E 6954509N

Sampled on 27 October 2020

This sample was collected from the 127.10 – 129.08 m depth interval of drillcore LSHD10, drilled in 1983 by Western Mining Corporation Limited at their Swanson Hill prospect (Brooke, 1984). The drillhole is located on Banjawarn Station, about 16.5 km north of Pinje-Eda Hill, 11.4 km northeast of Mason Hill, and 0.3 km east of the Urarey – Warren Bore Road.

Geological context

The unit sampled is a metadacite of the northern Duketon greenstone belt, near the western margin of the Burtville Terrane in the Eastern Goldfields Superterrane (Pawley et al., 2012). Samples of metadacite, collected from drillcore LSHD6, about 13.6 km to the south-southeast, yielded igneous crystallization ages of 2714 ± 5 Ma (GSWA 153317, Nelson, 2004a) and 2719 ± 5 Ma (GSWA 153318, Nelson, 2004b). Metadacite in drillcore LSHD7, about 11.3 km to the south, yielded an igneous crystallization age of 2716 ± 5 Ma (GSWA 235129, Wingate et al., 2021). Near the western margin of the Duketon belt, about 43 km to the south, a felsic volcaniclastic rock provided an igneous crystallization age of 2714 ± 4 Ma, and a porphyritic intrusion within that succession yielded an igneous crystallization age of 2704 ± 6 Ma (samples E380, E382, Kositcin et al., 2008).

Petrographic description

The sample is a medium-grained plagioclase-phyric metadacite (Fig. 1), consisting of about 65–70% groundmass, 25–30% plagioclase phenocrysts, 5% pseudomorphed phenocrysts, and accessory pyrite, pyrrhotite, magnetite, apatite and zircon. The groundmass is a granoblastic, microcrystalline quartzofeldspathic aggregate, overprinted by fine-grained metamorphic biotite, which is partially replaced by chlorite and by very fine-grained epidote. Biotite laths are subhedral to euhedral, about 0.05 mm long, and variably aligned parallel to plagioclase phenocrysts. Plagioclase phenocrysts are subhedral to euhedral, mostly tabular, up to 4 mm long, and partially altered to fine-grained chlorite, epidote and carbonate. Some plagioclase laths up to 1 mm long are aligned in a wavy foliation, which wraps around the margins of the largest phenocrysts. Pseudomorphed phenocrysts are tabular in form, up to 1.5 mm long, and are replaced by microcrystalline quartz, fine-grained biotite and minor carbonate. Disseminated subhedral pyrite and anhedral pyrrhotite are up to 0.5 mm in size. Magnetite is anhedral, mostly irregular, up to 0.3 mm in size, and locally enclosed by fine-grained biotite. Bladed apatite laths are up to 0.05 mm long. Although the sample has been metamorphosed at biotite grade, it is not obviously deformed, and the fabric is interpreted as an igneous flow foliation.

Zircon morphology

Zircons isolated from this sample are mostly colourless and subhedral. The crystals are up to 300 μm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 3–4 June 2021, using SHRIMP-B. Twelve analyses of the M257 standard were obtained during the session, of which 10 analyses indicated an external spot-to spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.15% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.24%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-five analyses were obtained from 25 zircons. Results are listed in Table 1, and shown in a concordia diagram (Figs 3, 4).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Three analyses are >5% discordant. The dates obtained from these three analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 24 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

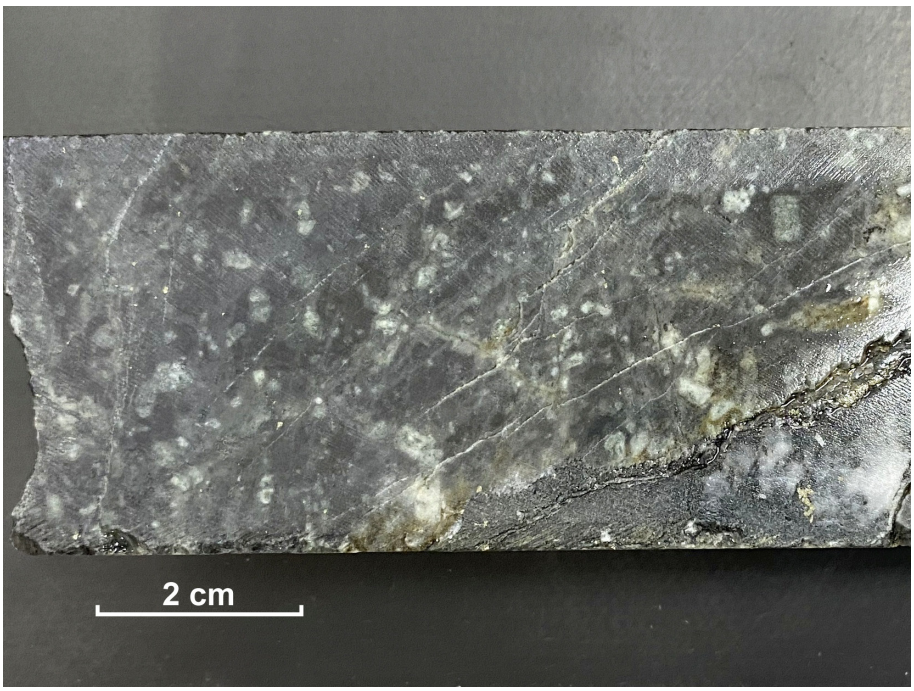

Figure 1. Drillcore image (wet surface) of sample 235199: metadacite, Duketon

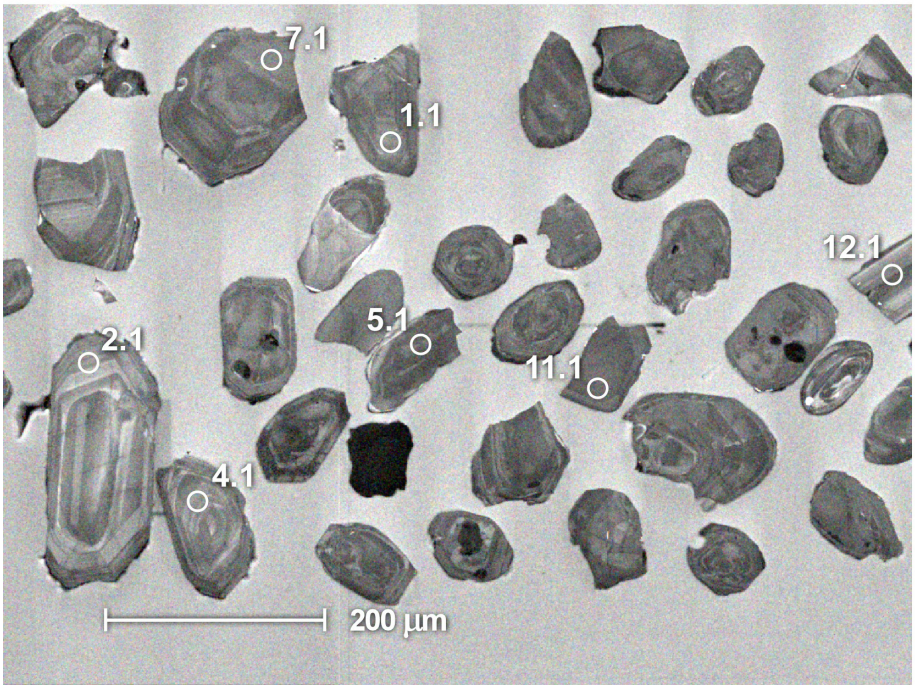

Figure 2. Cathodoluminescence image of representative zircons from sample 235199: metadacite, Duketon

Table 1. Ion microprobe analytical results for zircons from sample 235199: metadacite, Duketon

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	22	22.1	273	173	0.63	0.009	1.910	0.014	0.19655	0.00068	1.910	0.014	0.19646	0.00068	2714	17	2797	6	3.0
I	5	5.1	315	168	0.53	0.000	1.793	0.013	0.19657	0.00065	1.793	0.013	0.19657	0.00065	2857	17	2798	5	-2.1
I	9	9.1	204	89	0.44	0.006	1.883	0.015	0.19708	0.00078	1.883	0.015	0.19702	0.00078	2746	18	2802	7	2.0
I	24	24.1	164	83	0.50	-0.016	1.815	0.023	0.19710	0.00085	1.815	0.023	0.19724	0.00086	2830	30	2804	7	-0.9
I	11	11.1	219	148	0.68	0.011	1.873	0.015	0.19754	0.00074	1.873	0.015	0.19744	0.00075	2758	18	2805	6	1.7
I	19	19.1	324	206	0.64	0.022	1.884	0.014	0.19767	0.00062	1.885	0.014	0.19748	0.00063	2744	16	2805	5	2.2
I	6	6.1	243	120	0.49	0.021	1.865	0.015	0.19768	0.00073	1.865	0.015	0.19749	0.00074	2767	18	2806	6	1.4
I	18	18.1	186	77	0.42	0.007	1.865	0.015	0.19764	0.00079	1.865	0.015	0.19758	0.00079	2767	19	2806	7	1.4
I	16	16.1	223	160	0.72	-0.006	1.858	0.015	0.19767	0.00074	1.858	0.015	0.19772	0.00074	2776	18	2807	6	1.1
I	15	15.1	231	137	0.60	0.132	1.904	0.015	0.19909	0.00074	1.906	0.015	0.19791	0.00078	2718	17	2809	6	3.2
I	23	23.1	154	77	0.50	0.063	1.851	0.016	0.19849	0.00086	1.852	0.016	0.19793	0.00088	2783	20	2809	7	0.9
I	3	3.1	124	70	0.56	0.049	1.863	0.017	0.19844	0.00094	1.864	0.017	0.19799	0.00097	2769	21	2810	8	1.5
I	17	17.1	255	134	0.52	0.021	1.860	0.014	0.19829	0.00072	1.860	0.014	0.19811	0.00072	2773	17	2811	6	1.3
I	2	2.1	134	64	0.47	0.061	1.836	0.017	0.19888	0.00090	1.837	0.017	0.19833	0.00092	2801	21	2812	8	0.4
I	25	25.1	210	133	0.63	-0.012	1.838	0.015	0.19828	0.00076	1.838	0.015	0.19839	0.00076	2801	19	2813	6	0.4
I	21	21.1	248	162	0.65	0.005	1.851	0.014	0.19851	0.00072	1.851	0.014	0.19847	0.00072	2785	18	2814	6	1.0
I	1	1.1	137	65	0.47	-0.028	1.847	0.017	0.19830	0.00092	1.847	0.017	0.19855	0.00093	2790	21	2814	8	0.9
I	13	13.1	291	213	0.73	0.056	1.900	0.014	0.19916	0.00065	1.901	0.014	0.19866	0.00066	2725	17	2815	5	3.2
I	20	20.1	115	53	0.46	0.000	1.851	0.018	0.19872	0.00186	1.851	0.018	0.19872	0.00186	2784	22	2816	15	1.1
I	14	14.1	145	61	0.42	0.025	1.855	0.016	0.19905	0.00089	1.855	0.017	0.19883	0.00090	2779	20	2817	7	1.3
I	4	4.1	188	87	0.46	0.040	1.841	0.015	0.19992	0.00080	1.841	0.015	0.19957	0.00081	2796	19	2823	7	0.9
P	12	12.1	98	60	0.61	0.068	1.884	0.019	0.18951	0.00105	1.885	0.019	0.18890	0.00108	2743	23	2733	9	-0.4
D	7	7.1	498	352	0.71	0.020	3.553	0.024	0.14619	0.00059	3.554	0.024	0.14601	0.00060	1598	10	2300	7	30.5
D	10	10.1	467	343	0.74	0.007	2.499	0.017	0.17403	0.00063	2.499	0.017	0.17396	0.00063	2170	13	2596	6	16.4
D	8	8.1	447	383	0.86	0.041	2.558	0.018	0.17433	0.00088	2.559	0.018	0.17396	0.00088	2126	13	2596	8	18.1

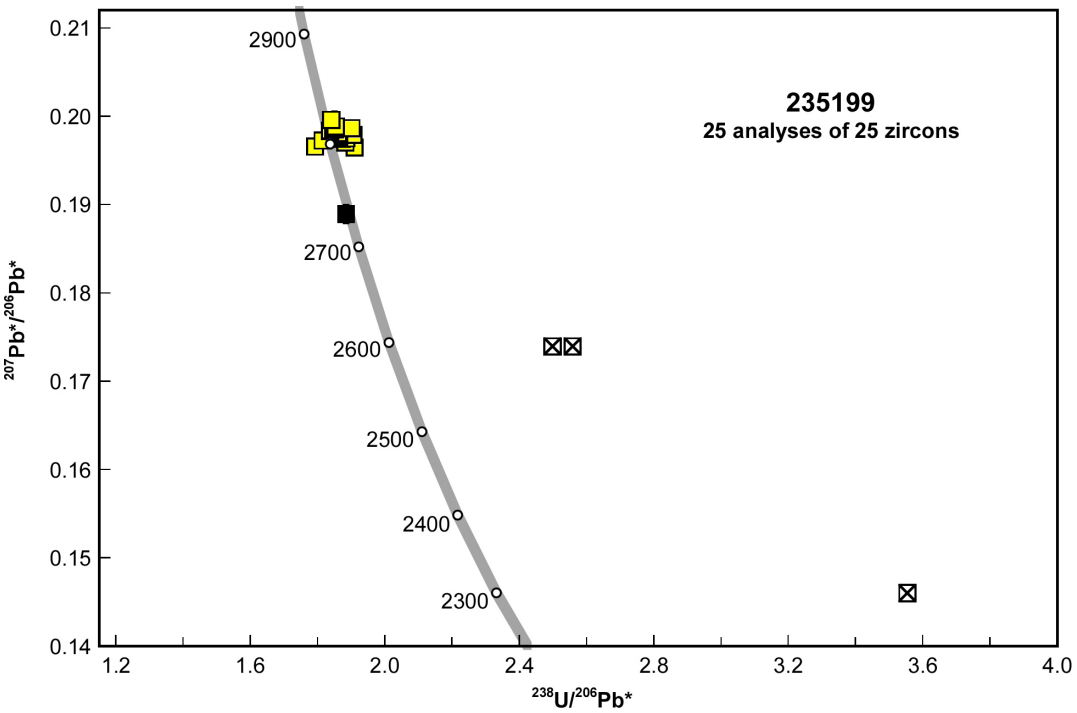

Figure 3. **U–Pb analytical data for sample 235199: metadacite, Duketon. Yellow squares indicate Group I (magmatic zircons); black square indicates Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%)**

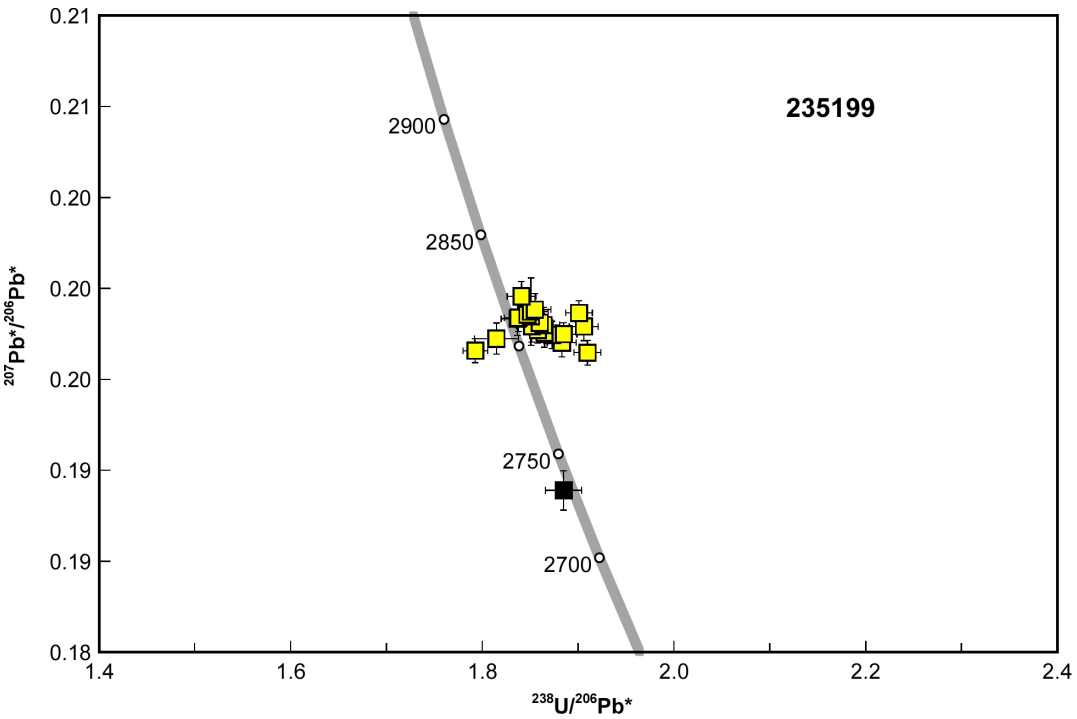

Figure 4. **Expanded view of U–Pb analytical data for zircons from sample 235199: metadacite, Duketon. Symbols as in Figure 3**

Group I comprises 21 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2808 ± 3 Ma (MSWD = 1.0).

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2733 ± 9 Ma (1σ).

The date of 2808 ± 3 Ma for the 21 analyses in Group I is interpreted as the igneous crystallization age of the dacite. The date of 2733 ± 9 Ma (1σ) for the single analysis in Group P is interpreted to reflect loss of radiogenic Pb.

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Grech, LL 2021, 235199: metadacite, Duketon; Geochronology Record 1834: Geological Survey of Western Australia, 5p.

Data obtained: 4 June 2021

Data released: 10 December 2021